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COMMENT GUIDELINES

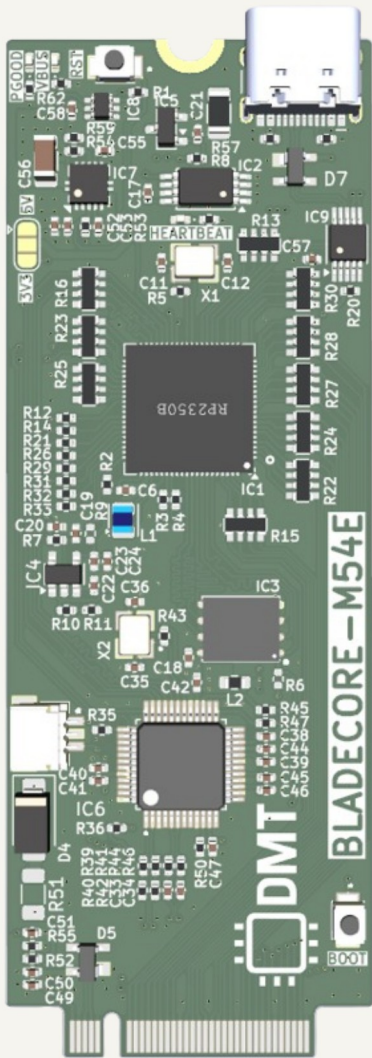
General comments are black, 50 mil size
Design notes and guidelines are blue, 50 mil size
Layout instructions are red, 50 mil size

NOTES

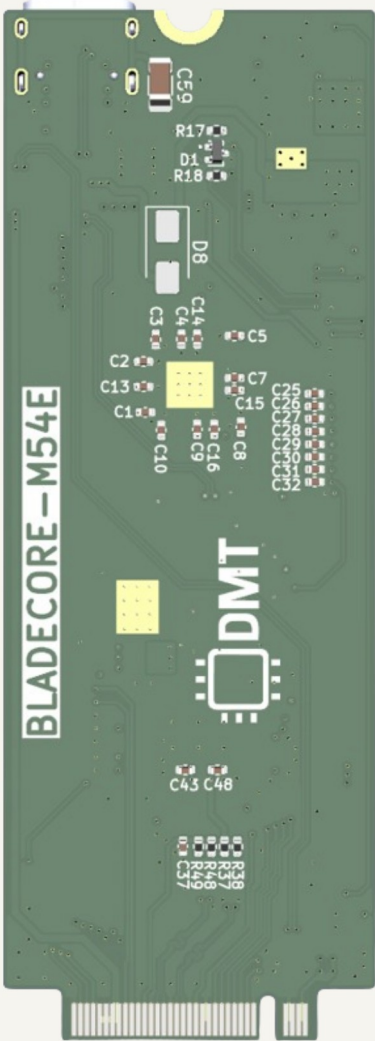
Not fitted components are marked as X

PCB PREVIEW

TOP VIEW



BOTTOM VIEW



 Date: 2026-01-31 Sheet Title: Root	Board Name: BladeCore		Project Name: M54	
	File Name: BladeCore-M54E.kicad_sch		Revision: 1.0.0	Variant: E
	Company: DvidMakesThings	Designer: David Sipos Reviewer:	Size: A3	Sheet: 1 of 9

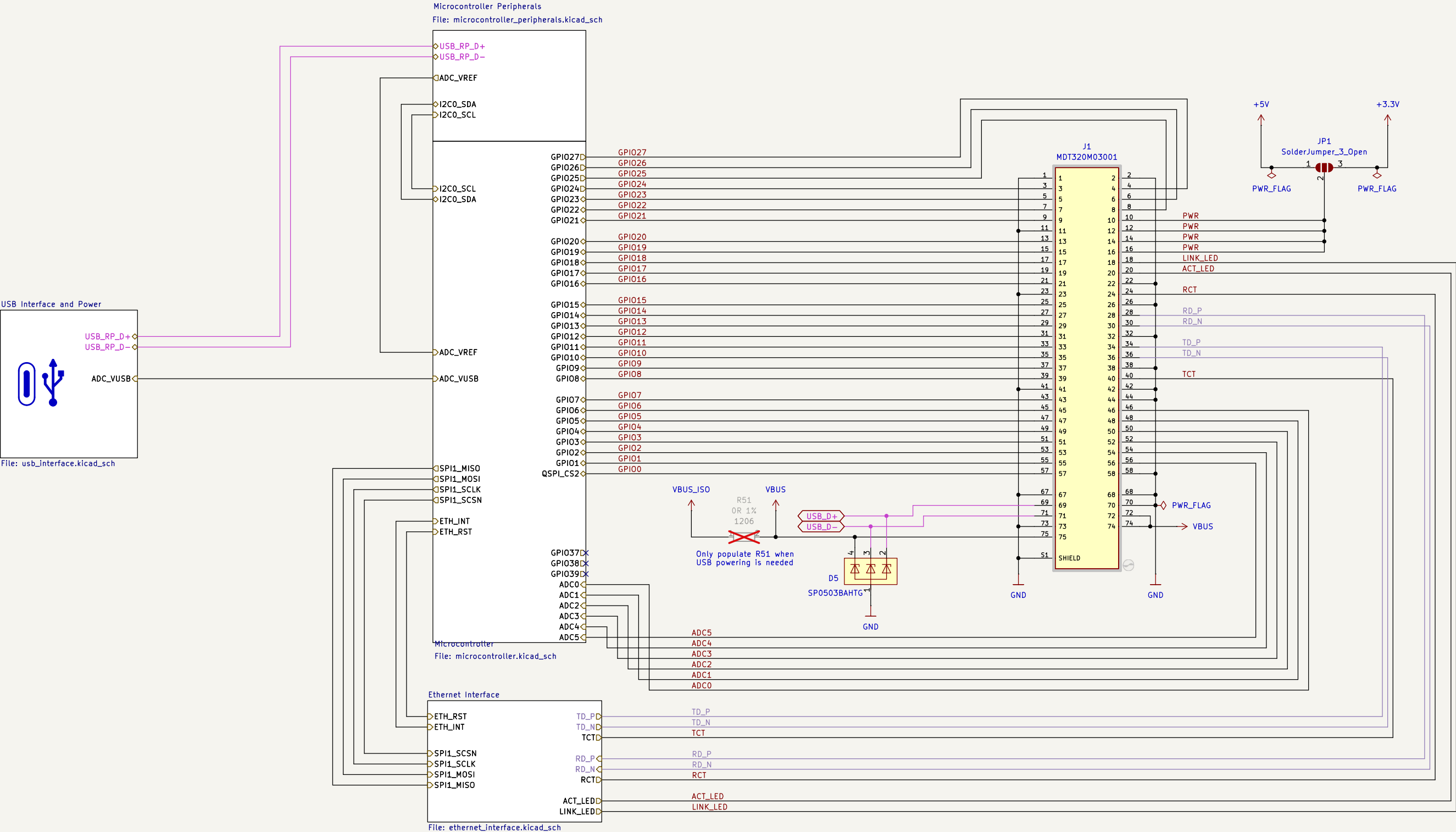
The diagram illustrates the internal architecture of the RP2354B SoC and its connections to various external components. The central component is the **RP2354B** SoC, which includes an **ONBOARD USB** controller, a **USB MUX**, and a **CARD EDGE USB** interface. The SoC is connected to a **2280 SIZE M.2 M-KEY CARD** via the **CARD EDGE USB** interface. The SoC also features several other interfaces and components:

- Power Management:** A **LDO** (Low Dropout Regulator) is connected to a **+5V** input and provides a **+3.3V** output. The **RP2354B** SoC is powered by **VBUS/3.3V/5V** and **+3.3V**.
- USB Interface:** The SoC is connected to a **USB INTERFACE** via the **ONBOARD USB** and **CARD EDGE USB** interfaces. The **USB INTERFACE** is connected to a **USB** port.
- GPIOs:** The SoC has several GPIO pins (GPIO0-27, GPIO28-31, GPIO32-33, GPIO36, GPIO40-45, GPIO46, GPIO47) connected to various external components.
- Peripherals:** The SoC includes an **ADC** (Analog-to-Digital Converter), **QSPI** (Quad Serial Peripheral Interface), **I2C0** (Inter-Integrated Circuit), **W5500 100BASE ETHERNET CONTROLLER**, **256KB EEPROM**, **W25Q128JVPIQ 8MB FLASH**, and a **HEARTBEAT LED**.
- External Components:** The SoC is connected to a **3.0V VREF** (Voltage Reference), a **VUSB** (USB Voltage), a **STATUS LEDS** (Status LEDs), and a **SWD HEADER** (Serial Wire Debug Header).

Input voltage:	3.3V - 5.5 V
Absolut maximum current consumption	480 mA
Typical current consumption	60mA
IO voltage	3.3V
Onboard LDO output capacity	1.5A
LDO Dropout	60mV

DMT Date: 2026-01-31	Board Name: BladeCore		Project Name: M54	
	File Name: Block Diagram.kicad_sch		Revision: 1.0.0	Variant: E
Sheet Title: Block Diagram	Company: DvidMakesThings	Designer: David Sipos Reviewer:	Size: A3	Sheet: 2 of 9

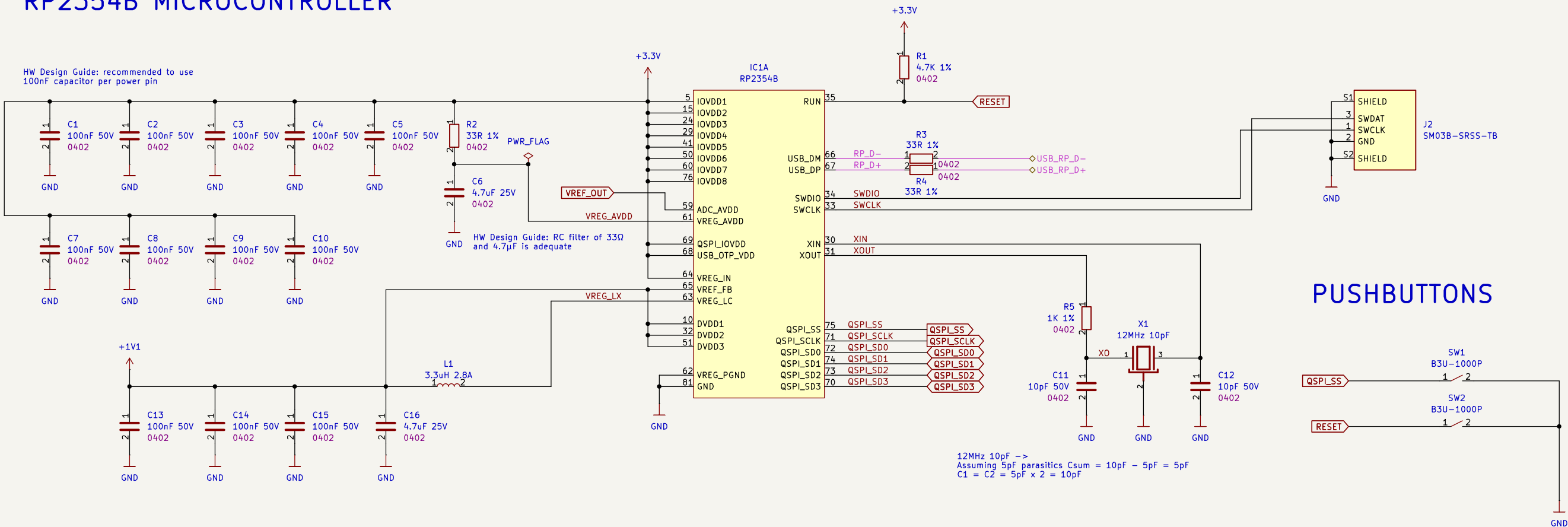
[3] Project Architecture



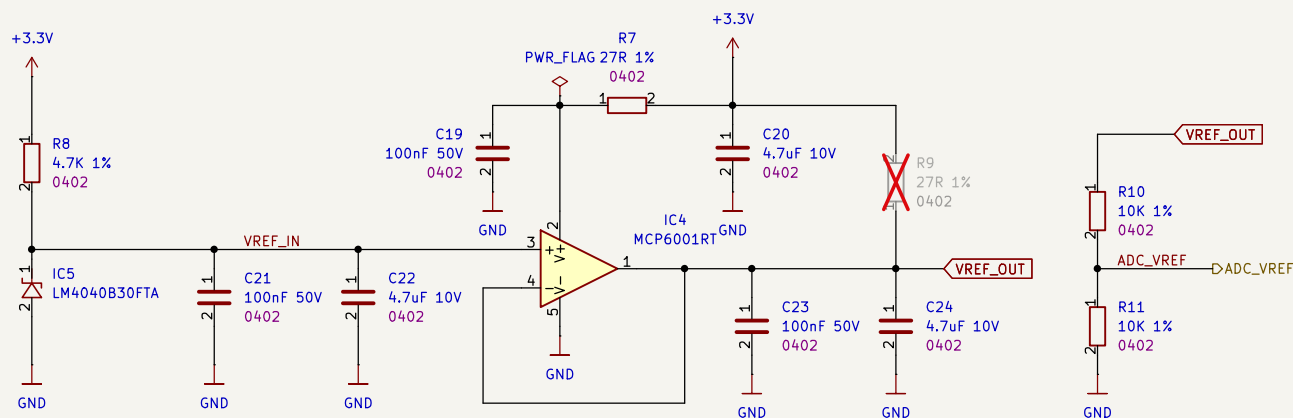
 Date: 2026-01-31 Sheet Title: Project Architecture	Board Name: BladeCore		Project Name: M54	
	File Name: Project Architecture.kicad_sch		Revision: 1.0.0	Variant: E
	Company: DvidMakesThings	Designer: David Sipos Reviewer:	Size: A3	Sheet: 3 of 9

[4] Microcontroller Peripherals

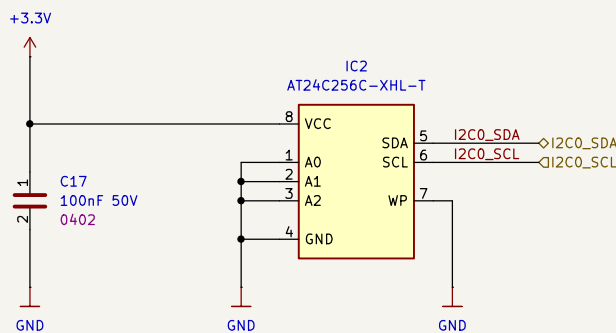
RP2354B MICROCONTROLLER



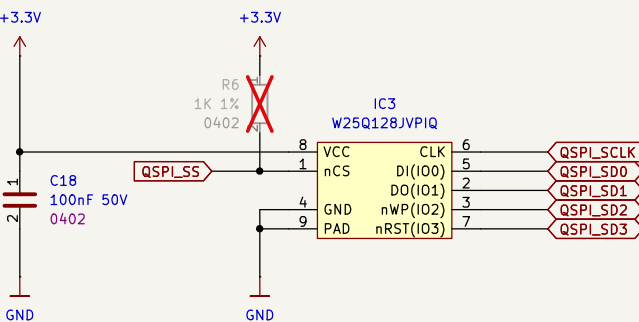
3.0V ADC VOLTAGE REFERENCE



EXTERNAL EEPROM



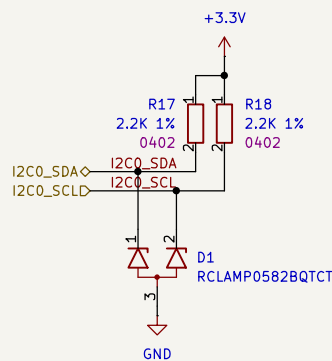
EXTERNAL FLASH



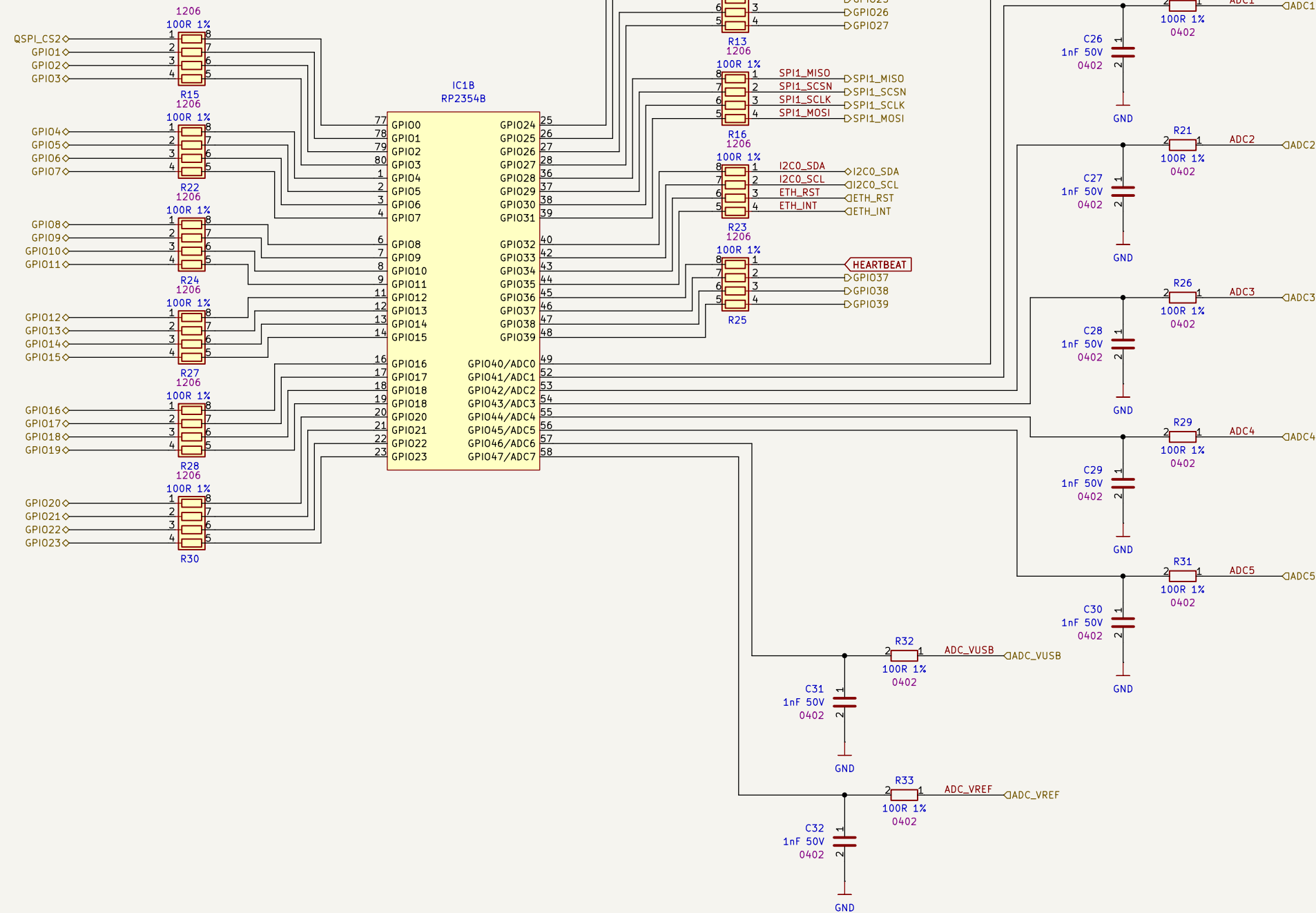
 Date: 2026-01-31 Sheet Title: Microcontroller Peripherals	Board Name: BladeCore		Project Name: M54	
	File Name: microcontroller_peripherals.kicad_sch		Revision: 1.0.0	Variant: E
	Company: DvidMakesThings	Designer: David Sipos Reviewer:	Size: A3	Sheet: 4 of 9

[5] Microcontroller

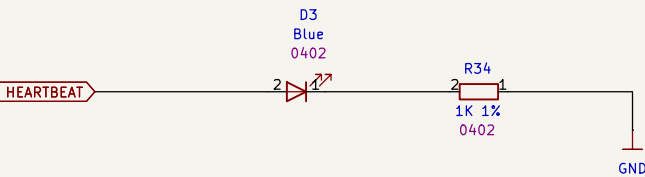
I2C PULLUP



RP2354B GPIO ASSIGNMENT



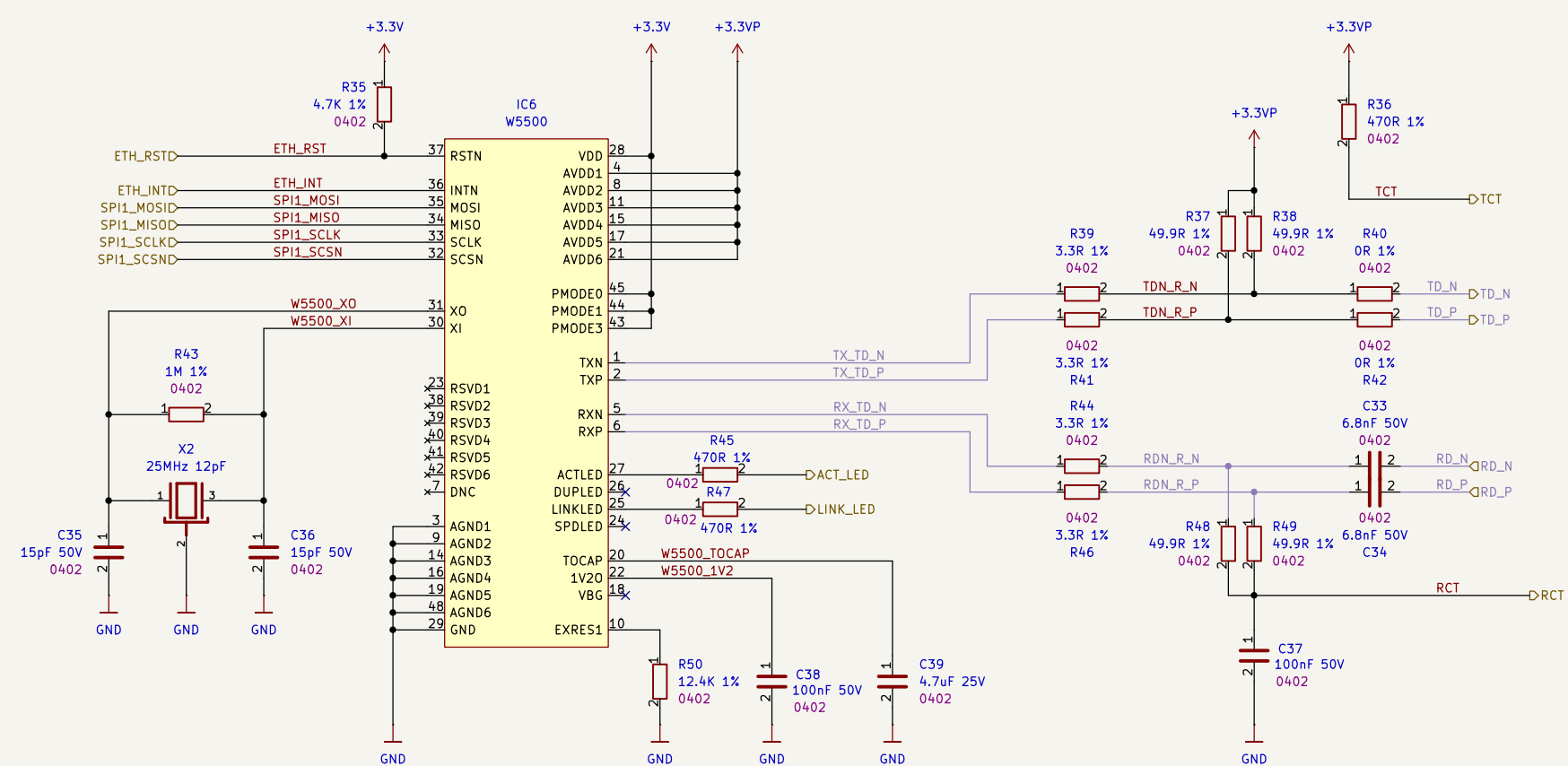
HEARTBEAT LED



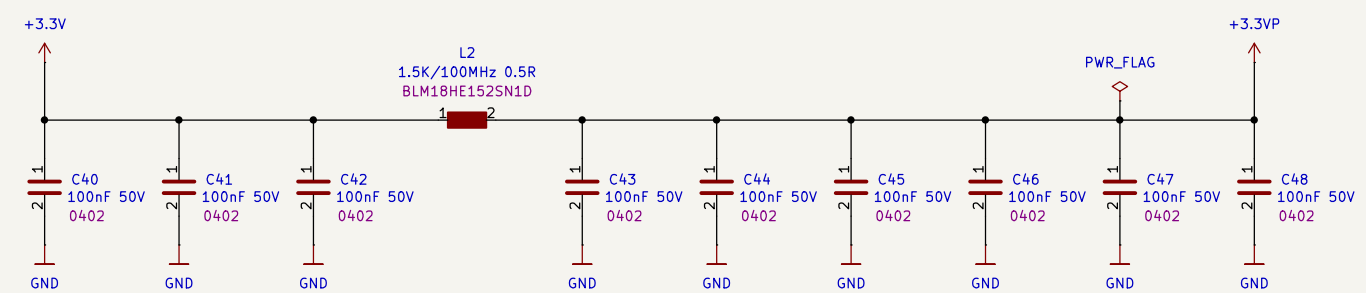
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	File Name: microcontroller.kicad_sch		Revision: 1.0.0	Variant: E
	Company: DvidMakesThings	Designer: David Sipos Reviewer:	Size: A3	Sheet: 5 of 9

[6] Ethernet Interface

W5500 ETHERNET CONTROLLER WITH PHY



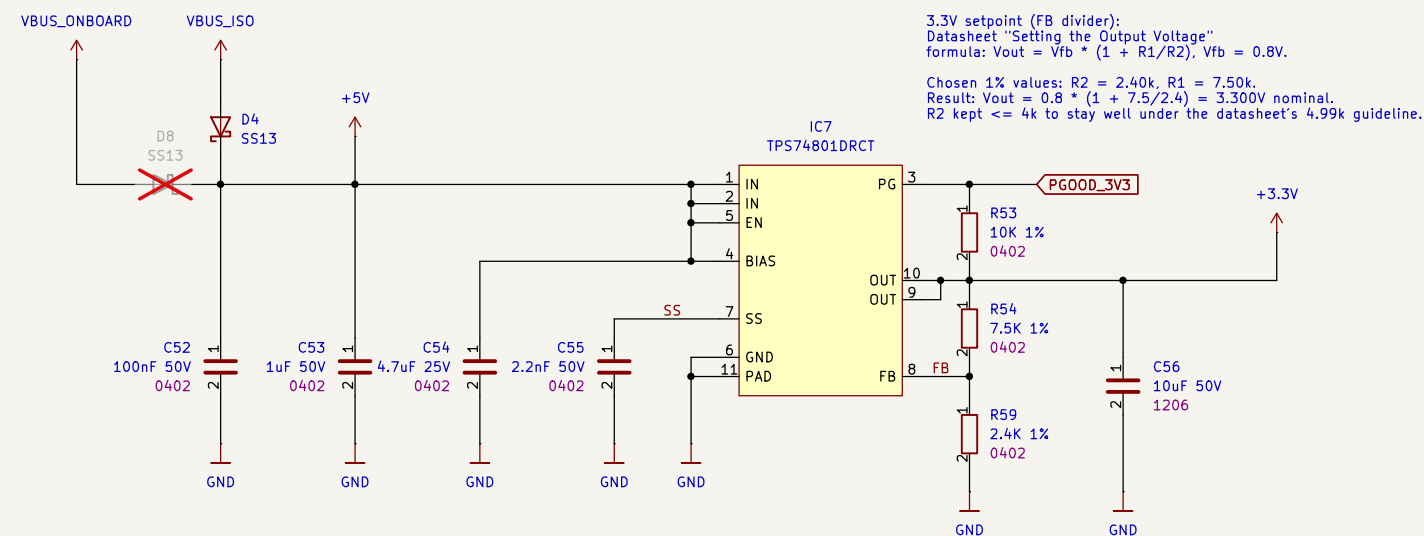
DECOUPLING



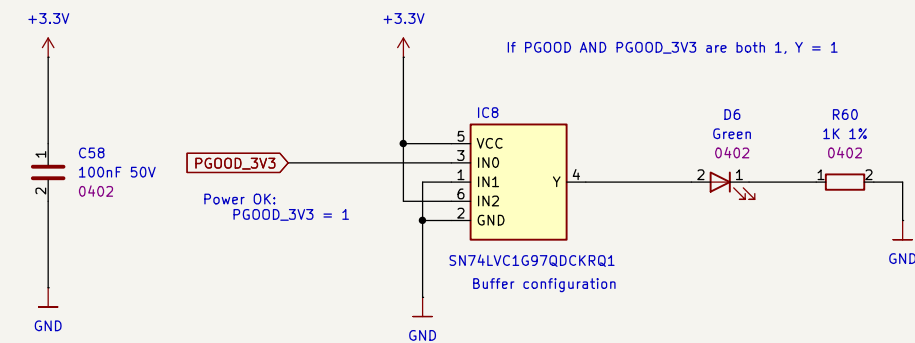
 Date: 2026-01-31 Sheet Title: Ethernet Interface	Board Name: BladeCore		Project Name: M54	
	File Name: ethernet_interface.kicad_sch		Revision: 1.0.0	Variant: E
	Company: DvidMakesThings	Designer: David Sipos Reviewer:	Size: A3	Sheet: 6 of 9

[7] USB Interface and Power

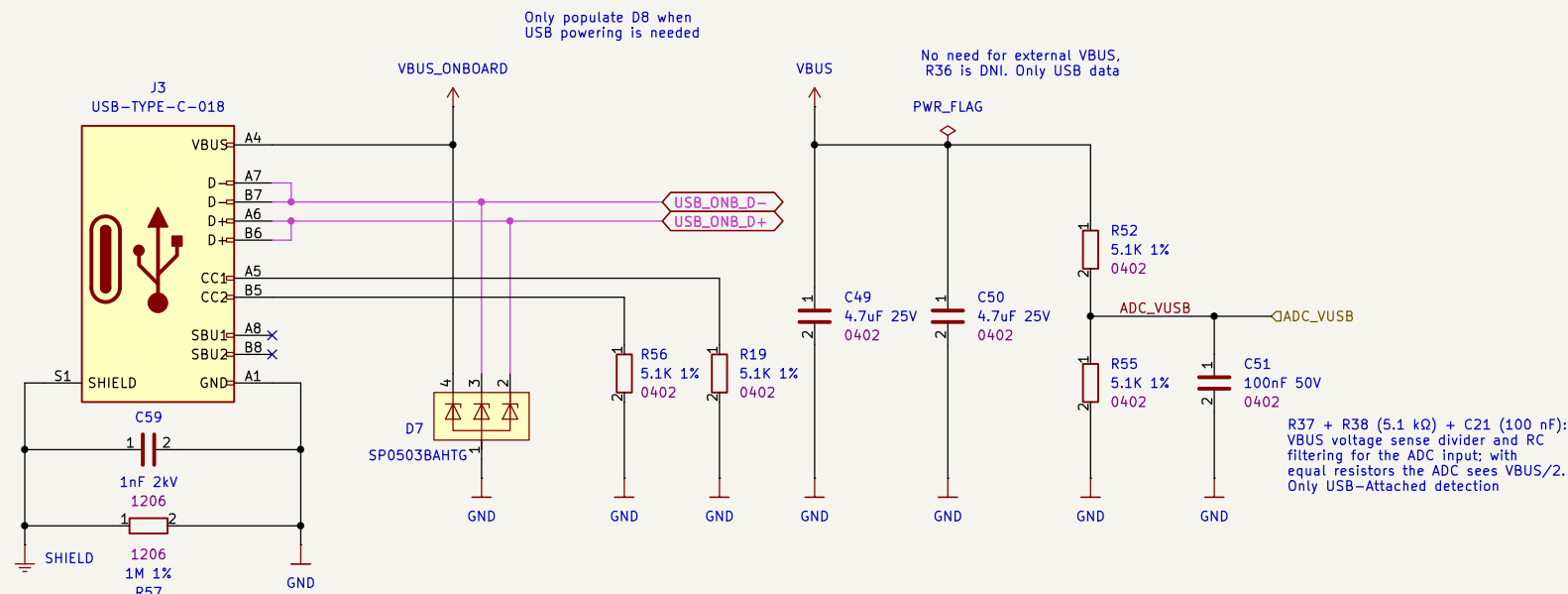
+5V to 3.3V PMIC



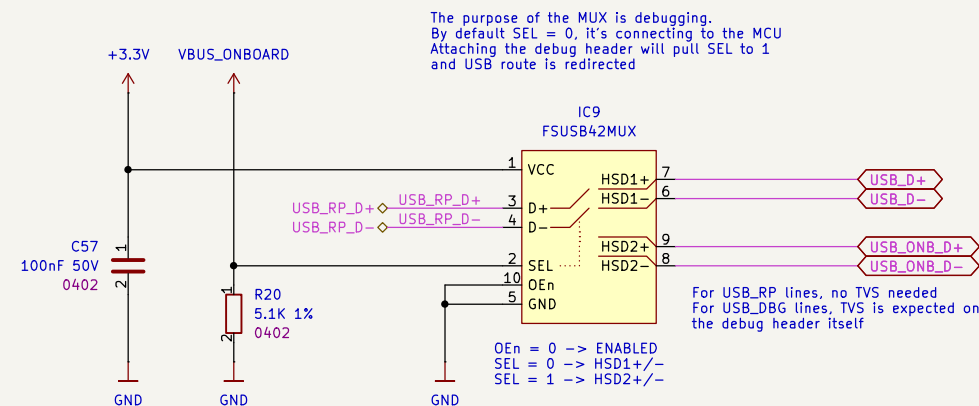
PGOOD LOGIC



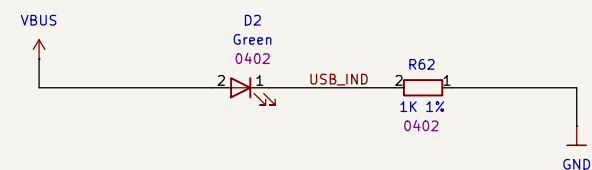
ONBOARD USB CONNECTOR



USB DATA-PATH SELECTOR

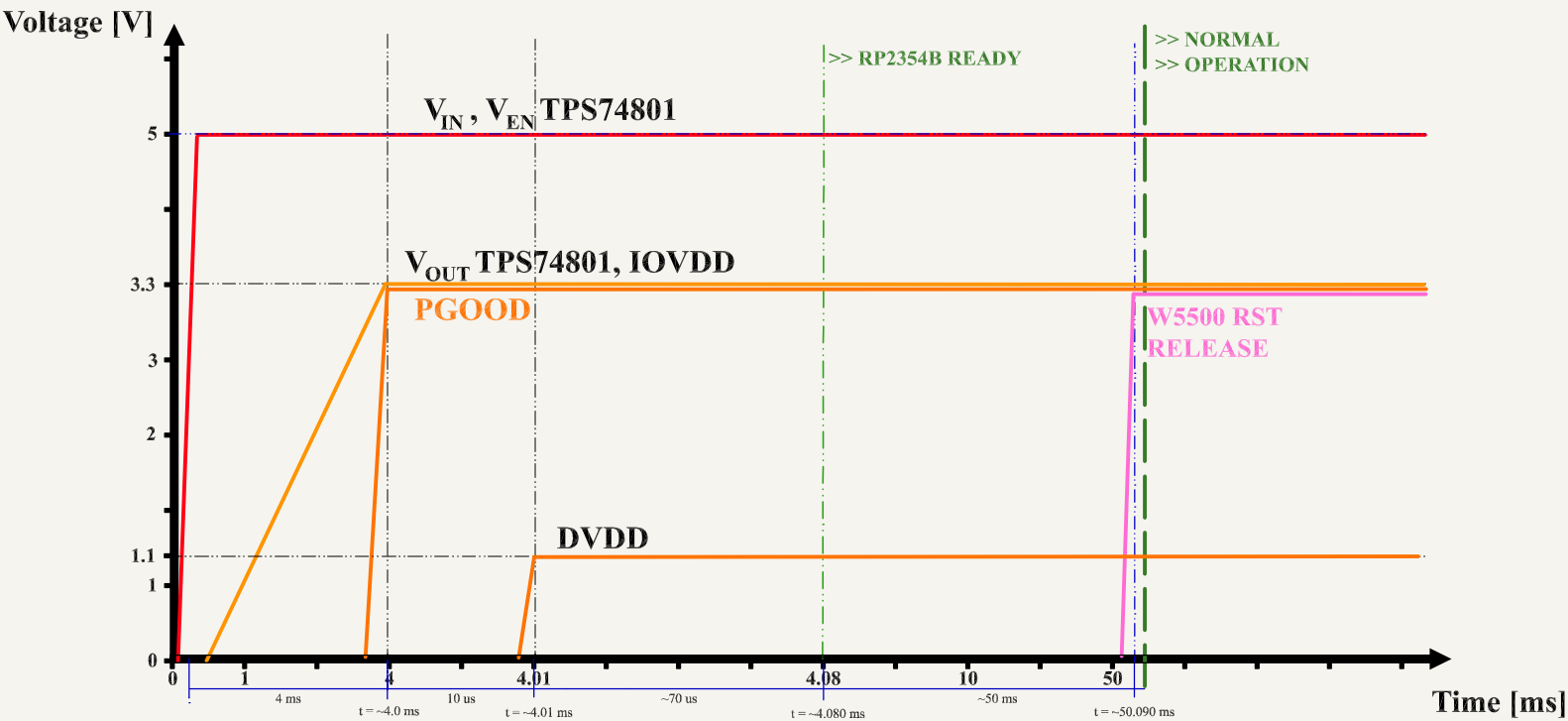


USB INDICATOR LED



 Date: 2026-01-31 Sheet Title: USB Interface and Power	Board Name: BladeCore		Project Name: M54	
	File Name: usb_interface.kicad_sch		Revision: 1.0.0	Variant: E
	Company: DvidMakesThings	Designer: David Sipos Reviewer:	Size: A3	Sheet: 7 of 9

[8] Power - Sequencing



#	Stage	Δt	Total elapsed
1	VIN applied (5.0V)	0	0.000 ms
2	VEN high (TPS74801 EN)	0	0.000 ms
3	VOUT = 3.3V (soft-start)	4.000 ms	4.000 ms
4	PGOOD valid	0	4.000 ms
5	RP2354B 3.3V present	0	4.000 ms
6	RP2354B 1.1V valid (POR)	0.010 ms	4.010 ms
7	RP2354B ready	0.080 ms	4.090 ms
8	W5500 ready	50.000 ms	54.090 ms

[9] Revision History

DATE	REVISION	RESPONSIBLE	CHANGE
31.01.2026	1.0.0	DMT	INITIAL CREATION

 Date: 2026-01-31	Board Name: BladeCore		Project Name: M54	
	File Name: Revision History.kicad_sch		Revision: 1.0.0	Variant: E
	Sheet Title: Revision History	Company: DvidMakesThings	Designer: David Sipos Reviewer:	Size: A3 Sheet: 9 of 9